

SOC Home Lab

Phishing Email Investigation & Threat Intelligence

PROJECT DOCUMENTATION REPORT

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Career track: SOC Analyst (entry-level)

Project type: Controlled phishing-email investigation

Lab Safety: The email, domains, and IP address used in this project are synthetic test data. No real recipient, production system, or real attacker infrastructure was targeted.

1. Executive Summary

This report documents a controlled phishing-email investigation designed to practice an entry-level SOC Analyst workflow. A synthetic email sample was created containing common phishing indicators, including brand impersonation, an urgent account-verification request, a suspicious-looking verification URL, mismatched sender/reply-to context, and failed SPF, DKIM, and DMARC authentication results.

The investigation extracted email, domain, URL, and IP indicators and assessed the available evidence without claiming a real-world compromise. The observed email is classified as a high-suspicion phishing sample for laboratory purposes and mapped to MITRE ATT&CK; T1566.002 — Phishing: Spearphishing Link.

Investigation Summary

Area	Result
Email Analysis	Completed
IOC Extraction	Completed
Authentication Analysis	SPF / DKIM / DMARC failures identified
Phishing Indicators	Multiple indicators identified
Threat Intelligence	Synthetic test indicators; no real malicious reputation claimed
MITRE ATT&CK	T1566.002 – Phishing: Spearphishing Link
Compromise Evidence	None collected

2. Project Objective

The objective was to investigate a phishing email as a SOC analyst would: identify suspicious characteristics, extract indicators of compromise (IOCs), assess authentication results, perform safe threat-intelligence lookups, map the behavior to MITRE ATT&CK;, and produce a documented analyst assessment.

3. Synthetic Email Sample

Indicator Type	Observed Value
Sender	security@micr0soft-security.example
Reply-To	account-verification@micr0soft-security.example
Domain	micr0soft-security.example
Verification URL	https://login-micr0soft-security.example/verify
Source IP	192.0.2.10
SPF	FAIL
DKIM	FAIL
DMARC	FAIL

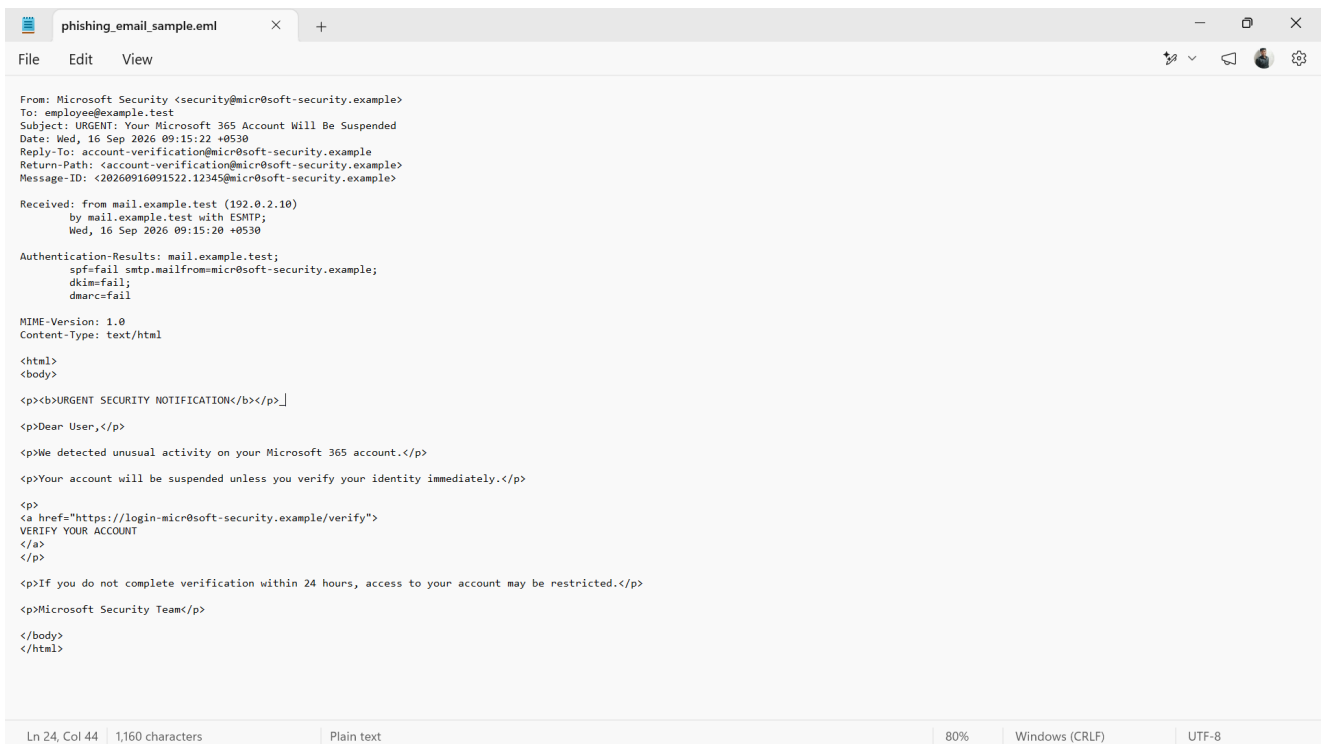
The `.example` domain and `192.0.2.10` address are used as synthetic documentation/test data. They are not evidence of a real malicious infrastructure.

4. Email Header Analysis

The header was reviewed for sender identity, routing information, reply handling, and email authentication results.

Key observations:

- The sender domain uses `micr0soft` with a zero in place of the letter `o`, creating a brand-impersonation indicator.
- The Reply-To address points to the same suspicious-looking domain but a different mailbox from the visible sender.
- The Received header identifies the synthetic source address `192.0.2.10`.
- SPF, DKIM, and DMARC are all shown as FAIL in the synthetic Authentication-Results header.
- The message uses urgent account-suspension language and asks the recipient to verify the account.



```
phishing_email_sample.eml
File Edit View

From: Microsoft Security <security@micr0soft-security.example>
To: employee@example.test
Subject: URGENT: Your Microsoft 365 Account Will Be Suspended
Date: Wed, 16 Sep 2026 09:15:22 +0530
Reply-To: account-verification@micr0soft-security.example
Return-Path: <account-verification@micr0soft-security.example>
Message-ID: <20260916091522.12345@micr0soft-security.example>

Received: from mail.example.test (192.0.2.10)
  by mail.example.test with ESMTP;
  Wed, 16 Sep 2026 09:15:20 +0530

Authentication-Results: mail.example.test;
  spf=fail smtp.mailfrom=micr0soft-security.example;
  dkim=fail;
  dmarc=fail

MIME-Version: 1.0
Content-Type: text/html

<html>
<body>

<p><b>URGENT SECURITY NOTIFICATION</b></p>

<p>Dear User,</p>

<p>We detected unusual activity on your Microsoft 365 account.</p>

<p>Your account will be suspended unless you verify your identity immediately.</p>

<p>
<a href="https://login-micr0soft-security.example/verify">
VERIFY YOUR ACCOUNT
</a>
</p>

<p>If you do not complete verification within 24 hours, access to your account may be restricted.</p>

<p>Microsoft Security Team</p>

</body>
</html>

Ln 24, Col 44 | 1,160 characters | Plain text | 80% | Windows (CRLF) | UTF-8
```

Figure 1. Synthetic phishing email header and body evidence.

5. IOC Extraction

The investigation extracted the following indicators from the sample:

- **Email:** security@microsoft-security.example
- **Reply-To:** account-verification@microsoft-security.example
- **Domain:** microsoft-security.example
- **URL:** https://login-microsoft-security.example/verify
- **IP:** 192.0.2.10

6. Threat Intelligence Assessment

The extracted domain, URL, and IP were treated as safe synthetic laboratory indicators. Because the sample deliberately uses the reserved `.example` domain and documentation IP range, a real malicious reputation was not expected and no real-world malicious reputation is claimed in this report.

The purpose of the threat-intelligence step was to demonstrate the SOC workflow of extracting an IOC, enriching it with reputation/context, and incorporating the result into an analyst assessment.

7. Phishing Risk Assessment

Indicator	Assessment
Brand impersonation	Suspicious
Sender / Reply-To context	Suspicious
SPF	FAIL
DKIM	FAIL
DMARC	FAIL
Urgent account suspension language	Suspicious
Credential / identity verification request	Suspicious
Verification URL	Suspicious-looking

8. MITRE ATT&CK; Mapping

Tactic	Technique	Sub-technique
Initial Access	T1566 – Phishing	T1566.002 – Spearphishing Link

Mapping rationale: the synthetic email contains a link presented as an account-verification action. The evidence supports mapping the phishing scenario to T1566.002. T1204.001 (User Execution: Malicious Link) is not asserted because there is no evidence in this lab that the link was clicked.

9. Analyst Assessment

FINAL CLASSIFICATION: PHISHING EMAIL — HIGH SUSPICION

The email contains multiple phishing indicators: brand impersonation, a suspicious verification URL, sender/reply-to concerns, failed SPF/DKIM/DMARC results, and urgent account-suspension language. The sample is mapped to T1566.002 — Phishing: Spearphishing Link.

No evidence was collected showing that the link was clicked, credentials were submitted, malware executed, or a successful compromise occurred.

10. Recommended SOC Response

1. Quarantine or flag the email according to the organization's email-security process.
2. Review the sender, Reply-To, domain, URL, and authentication results.
3. Search the environment for other messages containing the same indicators.
4. Determine whether any recipient interacted with the link or submitted credentials.
5. If interaction occurred, investigate the affected endpoint and account for follow-on activity.
6. Block confirmed malicious indicators according to organizational procedures.
7. Document the investigation and escalate according to incident-response policy.

11. SOC Workflow

Phishing Email → Header Analysis → IOC Extraction → Threat Intelligence → Risk Assessment → MITRE ATT&CK; Mapping → Analyst Assessment → Response

12. Skills Demonstrated

- Email Header Analysis
- Phishing Detection and Triage
- IOC Extraction
- Threat Intelligence Workflow
- URL and Domain Analysis
- Email Authentication Analysis
- MITRE ATT&CK; Mapping
- Incident Documentation
- SOC Analyst Investigation Workflow

13. Project Status

- Phishing Email Sample — Completed
- Email Header Analysis — Completed
- IOC Extraction — Completed
- Threat Intelligence Assessment — Completed
- Phishing Risk Assessment — Completed
- MITRE ATT&CK; Mapping — Completed
- Analyst Assessment — Completed
- SOC Response Recommendations — Completed
- Evidence Screenshot — Completed
- Documentation Report — Completed

14. Lab Disclaimer

This project uses synthetic phishing-email data for cybersecurity learning and SOC investigation practice. The `.example` domain and documentation IP address are non-production test indicators. No real person, organization, production system, or real attacker infrastructure was targeted.

15. Conclusion

This project demonstrates a practical phishing-email investigation workflow from initial email analysis through IOC extraction, threat-intelligence assessment, MITRE ATT&CK; mapping, analyst classification, and recommended SOC response. The work is intentionally framed as a controlled laboratory exercise and does not claim a real-world compromise.